LEVENTE NYIRI

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Education

Budapest University of Technology and Economics

Electrical Engineer B.Sc. 2021.09.01 - 2025.01.31

- ***Computer-Based Systems*** *specialization (AUT)*

Electrical Engineer M.Sc. 2025.02.01 - PRESENT

- ***Wireless Communication Systems*** *major (HVT)*
- ***FPGA-Based Systems*** *minor (MIT)*

Experience

AITIA International, Inc.

Hardware Design Engineer 2025.02 – 2025.08

- Schematic and **PCB design** – power/cost optimization, migrating from **Raspberry Pi** to **ESP32**
- **3D design** and **3D printing** – weatherproof enclosure design
- PCB assembly, mechanical assembly, and testing

IoT Intern 2024.06 – 2025.02

- Embedded software development for a **Raspberry Pi**-based system in **Python**

Relevant Projects

FPGA-Based audio compression system (M.Sc. thesis – in progress)

- Implementing an **RTL-level MP3 encoder** on a Xilinx Kintex-7 FPGA using **Verilog**.

Agricultural IoT: An embedded hardware solution (B.Sc. thesis)

- Awarded **2nd place at the TDK** (Scientific Students' Associations) conference.
- Developed firmware for on-site hardware units to monitor starling murmurations for **crop protection**.

"Persistence of vision" display

- Developed a persistence-of-vision (POV) display using an **STM32** microcontroller, shift registers, and a gyroscope to render arbitrary text and graphics via an LED array.
- Built a **C# .NET** application for it, where the user can draw and upload the graphic to be displayed.

Snake game for EFM32 Giant Gecko starter kit (ARM Cortex-M3)

- Developed bare-metal C drivers for LCD and peripherals.

RC car with camera

- Collaborated to develop an **ESP32-CAM-based** RC car. The view of the camera was streamed to a website.

Skills

C, Verilog, Bash, Python, Vivado, Vitis, Altium Designer, 3D Printing, Linux (Proficient), Git, English (C1)